

# CHAPTER-WISE PROBLEMS

Regression Analysis

Statistical Foundation of Data Science  
CSU1658

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## Contents

|          |                              |           |
|----------|------------------------------|-----------|
| <b>1</b> | <b>Linear Regression</b>     | <b>2</b>  |
| <b>2</b> | <b>Logistic Regression</b>   | <b>4</b>  |
| <b>3</b> | <b>Polynomial Regression</b> | <b>8</b>  |
| <b>4</b> | <b>Ridge Regression</b>      | <b>11</b> |

# 1 Linear Regression

## Problem 1: Simple Linear Regression with Hypothesis Testing

A researcher studies the relationship between study hours ( $x$ ) and exam scores ( $y$ ) for 6 students:

|     |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|
| $x$ | 2  | 4  | 5  | 7  | 9  | 11 |
| $y$ | 45 | 62 | 68 | 78 | 85 | 92 |

- (a) Compute the OLS estimates  $\hat{\beta}_0$  and  $\hat{\beta}_1$  using formulas:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- (b) Calculate the residuals  $e_i = y_i - \hat{y}_i$ , sum of squared errors (SSE), total sum of squares (SST), and coefficient of determination  $R^2$ .
- (c) Compute the residual standard error  $s = \sqrt{\text{SSE}/(n-2)}$  and test the null hypothesis  $H_0 : \beta_1 = 0$  at  $\alpha = 0.05$ . Use the test statistic:

$$t = \frac{\hat{\beta}_1 - 0}{s / \sqrt{\sum (x_i - \bar{x})^2}}$$

- (d) Construct a 95% confidence interval for  $\beta_1$ .

## Problem 2: Multiple Linear Regression Matrix Formulation

Consider a real estate dataset with house prices ( $y$ , in thousands) predicted by area ( $x_1$ , in 100 sq ft) and number of bedrooms ( $x_2$ ):

$$X = \begin{pmatrix} 1 & 12 & 2 \\ 1 & 15 & 3 \\ 1 & 18 & 3 \\ 1 & 20 & 4 \\ 1 & 25 & 4 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 180 \\ 220 \\ 260 \\ 280 \\ 340 \end{pmatrix}$$

- (a) Compute  $X^T X$  and  $X^T \mathbf{y}$ .
- (b) Calculate the OLS solution using the normal equations:

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$$

- (c) Compute the fitted values  $\hat{\mathbf{y}} = X \hat{\beta}$  and residuals  $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ .
- (d) Calculate  $R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$  where  $\text{SSE} = \mathbf{e}^T \mathbf{e}$  and  $\text{SST} = \sum (y_i - \bar{y})^2$ .

### Problem 3: Linear Regression with Categorical Predictors

A company analyzes employee salaries ( $y$ , in thousands) based on years of experience ( $x_1$ ) and department (categorical: Sales, Tech, HR). Data for 8 employees:

| Employee | Experience ( $x_1$ ) | Department | Salary ( $y$ ) |
|----------|----------------------|------------|----------------|
| 1        | 2                    | Sales      | 45             |
| 2        | 5                    | Sales      | 58             |
| 3        | 3                    | Tech       | 62             |
| 4        | 7                    | Tech       | 78             |
| 5        | 4                    | Tech       | 68             |
| 6        | 2                    | HR         | 42             |
| 7        | 6                    | HR         | 55             |
| 8        | 8                    | Sales      | 72             |

- (a) Create dummy variables for the department (use Sales as reference category):

$$D_{\text{Tech}} = \begin{cases} 1 & \text{if Tech} \\ 0 & \text{otherwise} \end{cases}, \quad D_{\text{HR}} = \begin{cases} 1 & \text{if HR} \\ 0 & \text{otherwise} \end{cases}$$

Write out the design matrix  $X$  (including intercept).

- (b) Set up and solve the normal equations to find  $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)^T$  where:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 D_{\text{Tech}} + \beta_3 D_{\text{HR}} + \epsilon$$

- (c) Interpret each coefficient: What does  $\hat{\beta}_2$  represent?  
 (d) Predict the salary for a Tech employee with 5 years of experience.

### Problem 4: Residual Analysis and Assumptions

Given the simple linear model  $y = \beta_0 + \beta_1 x + \epsilon$  with fitted values and residuals:

| $x$ | $y$  | $\hat{y}$ | $e$  |
|-----|------|-----------|------|
| 1   | 3.2  | 2.8       | 0.4  |
| 2   | 4.5  | 4.6       | -0.1 |
| 3   | 6.8  | 6.4       | 0.4  |
| 4   | 8.1  | 8.2       | -0.1 |
| 5   | 10.5 | 10.0      | 0.5  |
| 6   | 11.8 | 11.8      | 0.0  |
| 7   | 14.2 | 13.6      | 0.6  |
| 8   | 15.1 | 15.4      | -0.3 |

- (a) Compute the mean and variance of residuals. Verify that  $\sum e_i = 0$ .  
 (b) Calculate standardized residuals:  $r_i = e_i/s$  where  $s = \sqrt{\sum e_i^2 / (n-2)}$ .

- (c) Check for outliers: Are any  $|r_i| > 2$ ?
- (d) Compute the Durbin-Watson statistic to test for autocorrelation:

$$DW = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_{i=1}^n e_i^2}$$

Values near 2 indicate no autocorrelation. Interpret your result.

## Problem 5: Weighted Least Squares

When error variance is not constant ( $\text{Var}(\epsilon_i) = \sigma^2/w_i$ ), we use weighted least squares (WLS). Consider data with known weights:

| $x$ | $y$ | $w_i$ |
|-----|-----|-------|
| 1   | 2.5 | 1     |
| 2   | 3.8 | 2     |
| 3   | 5.5 | 2     |
| 4   | 6.8 | 4     |
| 5   | 8.2 | 4     |

- (a) The WLS estimator minimizes  $\sum w_i(y_i - \beta_0 - \beta_1 x_i)^2$ . Show that the WLS solution is:

$$\hat{\beta}_1^{WLS} = \frac{\sum w_i(x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sum w_i(x_i - \bar{x}_w)^2}$$

where  $\bar{x}_w = \sum w_i x_i / \sum w_i$ .

- (b) Compute  $\bar{x}_w$ ,  $\bar{y}_w$ , and the weighted sums needed for  $\hat{\beta}_1^{WLS}$  and  $\hat{\beta}_0^{WLS}$ .
- (c) Compare the WLS estimates with OLS (unweighted) estimates. Which should be preferred?
- (d) Calculate the weighted residuals  $e_i^w = \sqrt{w_i}(y_i - \hat{y}_i)$  and verify the WLS fit quality.

## 2 Logistic Regression

### Problem 6: Binary Logistic Regression Fundamentals

A credit card company models default probability based on debt-to-income ratio ( $x$ ). Data for 8 customers:

| Customer | Debt-to-Income ( $x$ ) | Default ( $y$ ) |
|----------|------------------------|-----------------|
| 1        | 0.15                   | 0               |
| 2        | 0.25                   | 0               |
| 3        | 0.35                   | 0               |
| 4        | 0.40                   | 1               |
| 5        | 0.50                   | 1               |
| 6        | 0.60                   | 1               |
| 7        | 0.70                   | 1               |
| 8        | 0.80                   | 1               |

(a) Write the logistic regression model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

where  $p = P(y = 1|x)$ . Express  $p$  in terms of the logit function.

(b) Suppose the estimated coefficients are  $\hat{\beta}_0 = -2.5$  and  $\hat{\beta}_1 = 8$ . Calculate the predicted probability of default for  $x = 0.45$ .

(c) Find the decision boundary: At what debt-to-income ratio is  $p = 0.5$ ?

(d) Compute the odds ratio for a 0.1 unit increase in debt-to-income ratio:

$$OR = \frac{p(x+0.1)/(1-p(x+0.1))}{p(x)/(1-p(x))} = e^{0.1\beta_1}$$

Interpret this value.

## Problem 7: Maximum Likelihood Estimation for Logistic Regression

For the logistic model with one predictor, the likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^n p_i^{y_i} (1-p_i)^{1-y_i}$$

where  $p_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$ .

Consider simplified data with 3 observations:

| $x$ | $y$ | Group |
|-----|-----|-------|
| 1   | 0   | A     |
| 2   | 1   | B     |
| 3   | 1   | C     |

- (a) Write the log-likelihood function:

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^n [y_i(\beta_0 + \beta_1 x_i) - \log(1 + e^{\beta_0 + \beta_1 x_i})]$$

- (b) Derive the gradient (score equations):

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^n (y_i - p_i), \quad \frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^n x_i (y_i - p_i)$$

- (c) For  $\beta_0 = 0$  and  $\beta_1 = 1$ , evaluate the log-likelihood  $\ell(0, 1)$  and the gradient components.
- (d) Explain why we need iterative methods (Newton-Raphson or gradient descent) to find MLEs.

### Problem 8: Multinomial Logistic Regression

A marketing study predicts customer preference (Low, Medium, High) based on price sensitivity ( $x$ ). Use High as the reference category. The model is:

$$\log \left( \frac{P(y = \text{Low})}{P(y = \text{High})} \right) = \beta_{01} + \beta_{11}x$$

$$\log \left( \frac{P(y = \text{Medium})}{P(y = \text{High})} \right) = \beta_{02} + \beta_{12}x$$

Given:  $\beta_{01} = 2$ ,  $\beta_{11} = -3$ ,  $\beta_{02} = 1$ ,  $\beta_{12} = -1.5$ .

- (a) For a customer with  $x = 0.5$ , compute the linear predictors:

$$\eta_1 = \beta_{01} + \beta_{11}(0.5), \quad \eta_2 = \beta_{02} + \beta_{12}(0.5)$$

- (b) Calculate the probabilities using the softmax function:

$$P(y = \text{Low}) = \frac{e^{\eta_1}}{1 + e^{\eta_1} + e^{\eta_2}}, \quad P(y = \text{Medium}) = \frac{e^{\eta_2}}{1 + e^{\eta_1} + e^{\eta_2}}, \quad P(y = \text{High}) = \frac{1}{1 + e^{\eta_1} + e^{\eta_2}}$$

- (c) Verify that probabilities sum to 1.
- (d) Which category has the highest predicted probability for this customer?

### Problem 9: Model Evaluation with Confusion Matrix

A logistic regression classifier predicts disease status (1 = diseased, 0 = healthy) for 10 patients. The threshold for classification is  $p \geq 0.5$ :

| Patient | True $y$ | Predicted $p$ | Predicted $\hat{y}$ |
|---------|----------|---------------|---------------------|
| 1       | 0        | 0.15          | 0                   |
| 2       | 0        | 0.30          | 0                   |
| 3       | 0        | 0.55          | 1                   |
| 4       | 1        | 0.45          | 0                   |
| 5       | 1        | 0.65          | 1                   |
| 6       | 1        | 0.75          | 1                   |
| 7       | 1        | 0.80          | 1                   |
| 8       | 0        | 0.20          | 0                   |
| 9       | 1        | 0.70          | 1                   |
| 10      | 0        | 0.40          | 0                   |

(a) Construct the confusion matrix:

|        |   | Predicted |      |
|--------|---|-----------|------|
|        |   | 0         | 1    |
| Actual | 0 | $TN$      | $FP$ |
|        | 1 | $FN$      | $TP$ |

(b) Calculate accuracy, precision, recall (sensitivity), and specificity:

$$\text{Accuracy} = \frac{TP + TN}{n}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}$$

(c) Compute the F1-score:

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

(d) If the threshold is changed to  $p \geq 0.6$ , how does the confusion matrix change? Discuss the trade-off between sensitivity and specificity.

### Problem 10: ROC Curve and AUC

For the data in Problem 9, we vary the classification threshold to generate the ROC curve.

(a) For thresholds  $\tau = 0.4, 0.5, 0.6, 0.7$ , compute the true positive rate (TPR) and false positive rate (FPR):

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

Create a table of (FPR, TPR) pairs.

- (b) Plot the ROC curve conceptually (sketch the points and connect them).
- (c) The area under the ROC curve (AUC) can be approximated using the trapezoidal rule. For two consecutive points  $(FPR_i, TPR_i)$  and  $(FPR_{i+1}, TPR_{i+1})$ :

$$\text{Area}_i = \frac{1}{2}(FPR_{i+1} - FPR_i)(TPR_i + TPR_{i+1})$$

Compute the total AUC.

- (d) Interpret the AUC value. What does  $\text{AUC} = 0.5$  represent? What about  $\text{AUC} = 1.0$ ?

### 3 Polynomial Regression

#### Problem 11: Quadratic Regression Fitting

A physicist models the height ( $y$ , in meters) of a projectile as a function of time ( $x$ , in seconds):

|     |   |     |     |     |     |     |
|-----|---|-----|-----|-----|-----|-----|
| $x$ | 0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 |
| $y$ | 0 | 11  | 18  | 21  | 20  | 15  |

Model:  $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$

- (a) Construct the design matrix  $X$ :

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_6 & x_6^2 \end{pmatrix}$$

- (b) Compute  $X^T X$  and  $X^T \mathbf{y}$ .
- (c) Solve the normal equations to find  $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$ .
- (d) Find the vertex of the parabola (maximum height):  $x^* = -\hat{\beta}_1 / (2\hat{\beta}_2)$ . Calculate the maximum height  $\hat{y}(x^*)$ .
- (e) Compute  $R^2$  for the quadratic model. Compare conceptually with a linear model (without  $x^2$  term).

#### Problem 12: Orthogonal Polynomials

When fitting high-degree polynomials, multicollinearity among  $x, x^2, x^3, \dots$  causes numerical instability. Orthogonal polynomials solve this.

Consider data:  $x = (-2, -1, 0, 1, 2)$ .

- (a) Show that the raw polynomial design matrix columns are correlated. Compute the correlation between  $x$  and  $x^2$ :

$$\text{cor}(x, x^2) = \frac{\sum (x_i - \bar{x})(x_i^2 - \bar{x^2})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (x_i^2 - \bar{x^2})^2}}$$

- (b) Construct orthogonal polynomials using Gram-Schmidt:

- $P_0(x) = 1$
- $P_1(x) = x - \bar{x}$
- $P_2(x) = x^2 - \langle x^2, P_0 \rangle P_0 - \langle x^2, P_1 \rangle P_1$

where  $\langle u, v \rangle = \sum u_i v_i / n$ .

- (c) Verify orthogonality:  $\sum P_1(x_i) P_2(x_i) = 0$ .
- (d) Explain why using orthogonal polynomials improves numerical stability in regression.

### Problem 13: Overfitting in Polynomial Regression

A data analyst fits polynomials of varying degrees to the data:

|     |     |     |     |     |      |
|-----|-----|-----|-----|-----|------|
| $x$ | 1   | 2   | 3   | 4   | 5    |
| $y$ | 2.1 | 3.9 | 6.2 | 7.8 | 10.1 |

- (a) For a degree-1 polynomial (linear), the fitted equation is  $\hat{y} = 0.1 + 2.0x$  with  $R^2 = 0.985$ .  
For a degree-4 polynomial,  $R^2 = 1.000$  (perfect fit).  
Compute the residual sum of squares (SSE) for both models. Which has smaller training error?
- (b) Calculate the adjusted  $R^2$  for both models:

$$R_{\text{adj}}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where  $p$  is the number of predictors. Which model is preferred by adjusted  $R^2$ ?

- (c) Explain the bias-variance trade-off: Why might the degree-4 polynomial perform poorly on new data despite perfect training fit?
- (d) Compute Akaike Information Criterion (AIC) for model selection:

$$AIC = n \log(\text{SSE}/n) + 2p$$

Use  $\text{SSE}_{\text{linear}} = 0.5$  and  $\text{SSE}_{\text{degree-4}} = 0.001$ . Which model is preferred by AIC?

### Problem 14: Polynomial Regression with Interaction Terms

A botanist studies plant growth ( $y$ ) as a function of sunlight hours ( $x_1$ ) and water amount ( $x_2$ ). The model includes a quadratic term and interaction:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_1 x_2 + \epsilon$$

Data for 5 plants:

| Plant | $x_1$ | $x_2$ | $y$ |
|-------|-------|-------|-----|
| 1     | 4     | 2     | 15  |
| 2     | 6     | 3     | 25  |
| 3     | 5     | 4     | 28  |
| 4     | 8     | 2     | 30  |
| 5     | 7     | 5     | 40  |

- Construct the design matrix  $X$  including all terms: intercept,  $x_1$ ,  $x_2$ ,  $x_1^2$ , and  $x_1 x_2$ .
- Given the fitted coefficients:  $\hat{\beta} = (2, 3, 2.5, -0.2, 0.5)^T$ , predict the growth for a plant with  $x_1 = 6$  and  $x_2 = 4$ .
- Interpret the interaction term  $\beta_4$ : How does the effect of sunlight on growth depend on water amount?
- Find the optimal sunlight hours (maximizing  $y$ ) for  $x_2 = 3$  by taking  $\frac{\partial y}{\partial x_1} = 0$  and solving:

$$\beta_1 + 2\beta_3 x_1 + \beta_4 x_2 = 0$$

### Problem 15: Spline Regression as Piecewise Polynomials

Spline regression fits different polynomials in different regions, connected smoothly at knots. Consider a cubic spline with one knot at  $x = 3$ :

$$f(x) = \begin{cases} \beta_{01} + \beta_{11}x + \beta_{21}x^2 + \beta_{31}x^3 & \text{if } x \leq 3 \\ \beta_{02} + \beta_{12}x + \beta_{22}x^2 + \beta_{32}x^3 & \text{if } x > 3 \end{cases}$$

For continuity and smoothness, we require:

- Continuity at  $x = 3$ :  $f(3^-) = f(3^+)$
- First derivative continuity:  $f'(3^-) = f'(3^+)$
- Second derivative continuity:  $f''(3^-) = f''(3^+)$

- Write the three continuity conditions in terms of the coefficients.
- Given  $\beta_{01} = 1$ ,  $\beta_{11} = 2$ ,  $\beta_{21} = -0.5$ ,  $\beta_{31} = 0.1$ , and assuming the constraints are satisfied, compute  $f(3)$  and  $f'(3)$ .
- Explain why splines are preferred over high-degree polynomials for flexible curve fitting.

- (d) In practice, splines are often represented using basis functions. Write the truncated power basis function for a cubic spline:

$$(x - c)_+^3 = \begin{cases} (x - c)^3 & \text{if } x > c \\ 0 & \text{otherwise} \end{cases}$$

How does this represent a knot at  $x = c$ ?

## 4 Ridge Regression

### Problem 16: Ridge Regression Derivation and Computation

Consider the housing price model with severe multicollinearity:

$$X = \begin{pmatrix} 1 & 1200 & 2.4 \\ 1 & 1800 & 3.6 \\ 1 & 2000 & 4.0 \\ 1 & 2500 & 5.0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 250 \\ 320 \\ 360 \\ 420 \end{pmatrix}$$

Note: The second and third columns are perfectly correlated ( $x_2 = 0.002x_1$ ).

- (a) Compute  $X^T X$  and show that it has a very small eigenvalue (near-singular), indicating multicollinearity. Calculate the condition number  $\kappa = \lambda_{\max}/\lambda_{\min}$ .
- (b) Derive the ridge regression estimator by minimizing:

$$\|\mathbf{y} - X\beta\|^2 + \lambda\|\beta\|^2$$

Show that the solution is:

$$\hat{\beta}^{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$$

- (c) For  $\lambda = 50$ , compute  $X^T X + 50I$  and find  $\hat{\beta}^{\text{ridge}}$ .
- (d) Compare with the OLS solution (if computable). Show that ridge shrinks coefficients toward zero.

### Problem 17: Bias-Variance Trade-off in Ridge Regression

The mean squared error (MSE) of an estimator decomposes as:

$$\text{MSE}(\hat{\beta}) = \text{Bias}^2(\hat{\beta}) + \text{Var}(\hat{\beta})$$

For ridge regression, the bias and variance are:

$$\text{Bias}(\hat{\beta}^{\text{ridge}}) = -\lambda(X^T X + \lambda I)^{-1} \beta$$

$$\text{Var}(\hat{\beta}^{\text{ridge}}) = \sigma^2 X(X^T X + \lambda I)^{-2} X^T$$

- (a) Show that as  $\lambda \rightarrow 0$ , ridge reduces to OLS:  $\hat{\beta}^{\text{ridge}} \rightarrow \hat{\beta}^{\text{OLS}}$ . What happens to bias and variance?
- (b) Show that as  $\lambda \rightarrow \infty$ ,  $\hat{\beta}^{\text{ridge}} \rightarrow \mathbf{0}$ . What happens to bias and variance?
- (c) For the simple case where  $X^T X = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$  (diagonal), show that the ridge shrinkage factor for the  $j$ -th coefficient is:

$$\frac{\lambda_j}{\lambda_j + \lambda}$$

- (d) Explain intuitively: Why does ridge regression improve prediction accuracy despite introducing bias?

### Problem 18: Cross-Validation for Lambda Selection

To choose the optimal regularization parameter  $\lambda$ , we use cross-validation. Consider Leave-One-Out CV (LOOCV):

$$\text{Data: } X = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 8 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} 3 \\ 7 \\ 11 \\ 15 \end{pmatrix}$$

- (a) For  $\lambda = 1$ , compute the ridge coefficients  $\hat{\beta}^{\text{ridge}}$  using all data.
- (b) For LOOCV, we leave out observation  $i$  and compute  $\hat{y}_{-i}$  (prediction for  $i$  using model trained on remaining data). The LOOCV shortcut formula is:

$$\hat{y}_i - \hat{y}_{-i} = \frac{e_i}{1 - H_{ii}}$$

where  $H = X(X^T X + \lambda I)^{-1} X^T$  is the hat matrix and  $e_i = y_i - \hat{y}_i$ .

Compute the hat matrix  $H$  for  $\lambda = 1$ .

- (c) Calculate the LOOCV error:

$$CV(\lambda) = \frac{1}{n} \sum_{i=1}^n \left( \frac{e_i}{1 - H_{ii}} \right)^2$$

- (d) Repeat for  $\lambda = 0.5$  and  $\lambda = 2$ . Which  $\lambda$  minimizes CV error?

### Problem 19: Ridge Regression Path and Degrees of Freedom

The ridge trace plots coefficient paths  $\hat{\beta}_j(\lambda)$  as  $\lambda$  varies. The effective degrees of freedom is:

$$df(\lambda) = \text{tr}(H_\lambda) = \text{tr}(X(X^T X + \lambda I)^{-1} X^T)$$

For the dataset:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

- (a) Compute  $\hat{\beta}^{\text{ridge}}$  for  $\lambda = 0, 1, 5, 10$ .
- (b) For each  $\lambda$ , calculate  $df(\lambda) = \text{tr}(H_\lambda)$ . Verify that  $df(0) = p$  (number of predictors) and  $df(\infty) \rightarrow 0$ .
- (c) Plot the ridge path conceptually: Sketch how  $\hat{\beta}_1$  and  $\hat{\beta}_2$  change as  $\lambda$  increases from 0 to 10.
- (d) The generalized cross-validation (GCV) criterion is:

$$GCV(\lambda) = \frac{\sum e_i^2}{(1 - df(\lambda)/n)^2}$$

For  $\lambda = 1$ , compute GCV if  $\sum e_i^2 = 0.5$  and  $df(1) = 1.5$ .

## Problem 20: Ridge Regression vs LASSO

Compare ridge regression with LASSO (Least Absolute Shrinkage and Selection Operator):

**Ridge:**  $\min_{\beta} \|\mathbf{y} - X\beta\|^2 + \lambda\|\beta\|_2^2$

**LASSO:**  $\min_{\beta} \|\mathbf{y} - X\beta\|^2 + \lambda\|\beta\|_1$

Consider a simple case with two predictors and OLS estimates  $\hat{\beta}^{\text{OLS}} = (6, 8)^T$ .

- (a) Sketch the constraint regions in  $(\beta_1, \beta_2)$  space:

- Ridge:  $\beta_1^2 + \beta_2^2 \leq t$  (circle)
- LASSO:  $|\beta_1| + |\beta_2| \leq t$  (diamond)

Also sketch the OLS contours (ellipses centered at  $\hat{\beta}^{\text{OLS}}$ ).

- (b) Explain geometrically why LASSO can produce exact zeros (sparse solutions) while ridge only shrinks toward zero.
- (c) For the constraint  $\beta_1^2 + \beta_2^2 = 25$ , if ridge yields  $\hat{\beta}^{\text{ridge}} = (3, 4)^T$ , find the corresponding  $\lambda$  by solving:

$$\hat{\beta}^{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$$

(Assume  $X^T X = I$  for simplicity and  $X^T \mathbf{y} = (6, 8)^T$ ).

- (d) Discuss when to use ridge vs LASSO: Which is better for prediction? For interpretation?